Atmospheric entry

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use energy and geometry to discuss entry heating without claiming a guaranteed outcome.

SUCCESS CRITERIA - I can...

1. Compare kinetic energy with a stated heating requirement.
2. Explain the role of surface area relative to mass.
3. Name missing factors in a survival prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A process has ΔU=-15 J and W_{by}=21 J. Find Q and decide whether heat enters or leaves.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $K = \frac{1}{2}mv^2$

$Q_{absorbed}$ = fraction absorbed × energy available.

Entry also involves ablation and fragmentation. An energy comparison alone cannot predict survival.

Worked reasoning: A fictional entering object loses 1000 J of mechanical energy; a model assigns 700 J to heated air and 300 J to the object. Those shares are supplied assumptions, not universal atmospheric-entry fractions.

Your turn: Draw the energy budget and identify one measurement needed to test the shares.

Marzano check - circle: ① ② ③

03 Represent and interpret

Compare a small and a large sphere of the same material. Which has more surface area per unit mass? What does that suggest about surface interactions per kilogram?



A B

Marzano check - circle: ① ② ③

04 Solve and justify

For a hypothetical 10 kg rock at 15 km/s, compute K. A supplied heating/vaporization requirement is 8.6×10^6 J/kg. Find total requirement and energy ratio.

Marzano check - circle: ① ② ③

05 Analyze the reasoning

Name at least three additional quantities needed before claiming a particular rock completely ablates or reaches the ground.

Marzano check - circle: 1 2 3

06 Choose a method

For fixed GM, launch radius becomes 16 times as large. Compare ideal escape speeds and name the equation.

Marzano check - circle: 1 2 3

07 Exit check

If radius doubles at fixed density, derive the area-to-mass multiplier and explain why kinetic-energy-per-mass alone cannot predict a size threshold for survival.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 2 kg sample has 20 kJ deposited as internal energy. With effective c=500 J/(kg·K), estimate ΔT if no phase change or loss occurs. State why this is not a survival threshold.

Marzano check - circle: 1 2 3